

# StoryNest

## An AI-Powered Bilingual (English + Hindi) Reading & Literacy Platform

Creating personalized stories, interactive narration, and gamified comprehension experiences for children (Ages 3–12).

■ **Google Gemini AI**

Generates personalized bilingual stories + vocabulary + quizzes atomically.

■ **Bilingual Literacy**

English + Hindi text side-by-side with synchronized Web Speech TTS narration.

■ **Gamified Quizzes**

Comprehension checks, star rewards, daily streaks & printable certificates.

■ **Dual Dashboards**

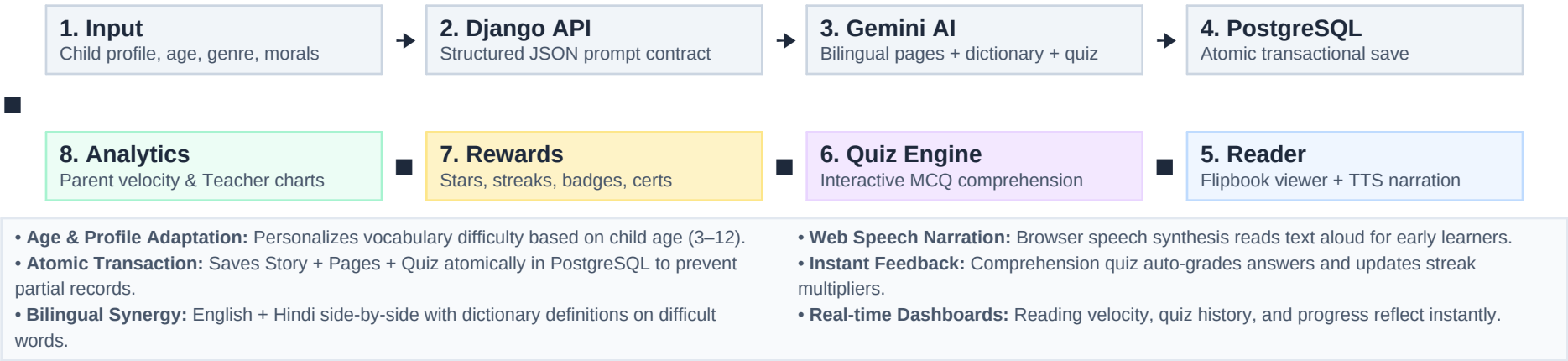
Dedicated Parent & Teacher portals with learning velocity & reading analytics.

■ **Live Explanation Guide:**

*“StoryNest ek AI-powered educational platform hai jo children ke liye personalized English-Hindi stories generate karta hai aur interactive reading, quizzes aur gamification ke through literacy improve karta hai.”*

# Slide 2 — Core Learning & AI Architecture Flow

Visual step-by-step pipeline from child personalization to gamification and analytics



# Slide 3 — User Roles, Permissions & Security Architecture

Strict role separation and row-level ownership validation across all portals

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| <div>■■■■■ PARENT PORTAL</div> <ul style="list-style-type: none"><li>• Manage child profiles &amp; interests</li><li>• Generate &amp; read custom stories</li><li>• Track daily streaks, goals &amp; bedtime schedule</li><li>• View AI vocabulary growth &amp; progress reports</li><li>• Download milestone achievement certificates</li></ul>                                                                                                                                                                                                        | <div>■■■ TEACHER HUB</div> <ul style="list-style-type: none"><li>• Classroom management &amp; unique join codes</li><li>• Student rosters &amp; individual skill breakdowns</li><li>• Story homework assignments with due dates</li><li>• Quiz submissions grading &amp; pass rate metrics</li><li>• Automated academic PDF report cards</li></ul> | <div>■■ ADMIN SUITE</div> <ul style="list-style-type: none"><li>• System-wide user directory &amp; role controls</li><li>• Overall platform usage analytics &amp; story counts</li><li>• System health monitoring &amp; active logs</li><li>• Content moderation &amp; database management</li></ul> |
| <div><div>■ Verified Row-Level Data Isolation:</div><ul style="list-style-type: none"><li>• Parent A → Child A Data: HTTP 200 OK (Authorized)</li><li>• Parent A → Child B Data: HTTP 404 NOT FOUND (Access Denied)</li><li>• Parent B → Child A Data: HTTP 404 NOT FOUND (Protected)</li></ul></div> <div><div>■ Live Explanation Note:</div><p><i>"Backend ownership validation filters ensure karte hain ki changing an object ID in API requests will never leak or expose another parent's children, reading logs, or private data."</i></p></div> |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                      |

# Slide 4 — Technology Architecture & Cloud Deployment

Production-grade decoupled stack with high performance and security guarantees

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| <div>■ FRONTEND CLIENT</div> <ul style="list-style-type: none"><li>• React 19 + Vite 8</li><li>• Tailwind CSS 4 Glassmorphism</li><li>• Axios with JWT Interceptors</li><li>• Framer Motion Flipbook Reader</li><li>• jsPDF &amp; html2canvas</li><li>• Hosted on: Netlify / Vercel</li></ul>                                                                                                                                                        | <div>■■ BACKEND API</div> <ul style="list-style-type: none"><li>• Python 3.12 / Django 5</li><li>• Django REST Framework</li><li>• SimpleJWT Token Rotation</li><li>• Atomic Transactions</li><li>• WhiteNoise Static Assets</li><li>• Hosted on: Render / AWS</li></ul> | <div>■■ DATABASE &amp; AI SERVICES</div> <ul style="list-style-type: none"><li>• PostgreSQL Cloud Database</li><li>• Google Gemini 1.5 / 2.0 API</li><li>• Web Speech Audio API</li><li>• Environment Key Isolation</li><li>• CORS Whitelisting</li><li>• Storage: Managed DB</li></ul> |
| <div><div>■ JWT Authentication: Access + Refresh tokens with expiry</div><div>■ Role Authorization: Route guards on Parent, Teacher, Admin</div><div>■ Ownership Validation: Child/Story access scoped strictly to user</div><div>■ Zero Secret Exposure: Gemini &amp; DB keys strictly in .env</div><div>■ Production CORS: Whitelisted domain origins only</div><div>■ Clean Builds: 0 Vite compile errors &amp; 0 Django check issues</div></div> |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                         |

# Slide 5 — Developer Golden Rules & Project Status

Standards for maintaining stability, scalability, and seamless integration

## ■ PARENT MODULE STATUS: 100% COMPLETE, VERIFIED & PRODUCTION READY

### 1. Preserve Existing Contracts

Never break or rework working Parent & Teacher logic or existing API response formats.

### 3. Security Testing Mandatory

Always execute cross-user ID tampering tests (asserting HTTP 404 on unauthorized data).

### 2. Instant Database Migrations

Any model modification must be followed immediately by `makemigrations` and `migrate`.

### 4. Full-Stack Verification

Always run ``npm run build`` and ``python manage.py check`` before committing changes.

### ■ *Final Presentation Conclusion:*

*"StoryNest is not just an AI story generator; it is a complete bilingual literacy ecosystem that combines personalized storytelling, interactive reading, assessment, gamification, and learning analytics in one secure, production-ready platform."*